# Sentiment-Augmented Trading System: FinFlow 3.0

Live Dashboard

Od Sukh-Ochir, Department of Data Science, Hanyang University

## Research Motivations

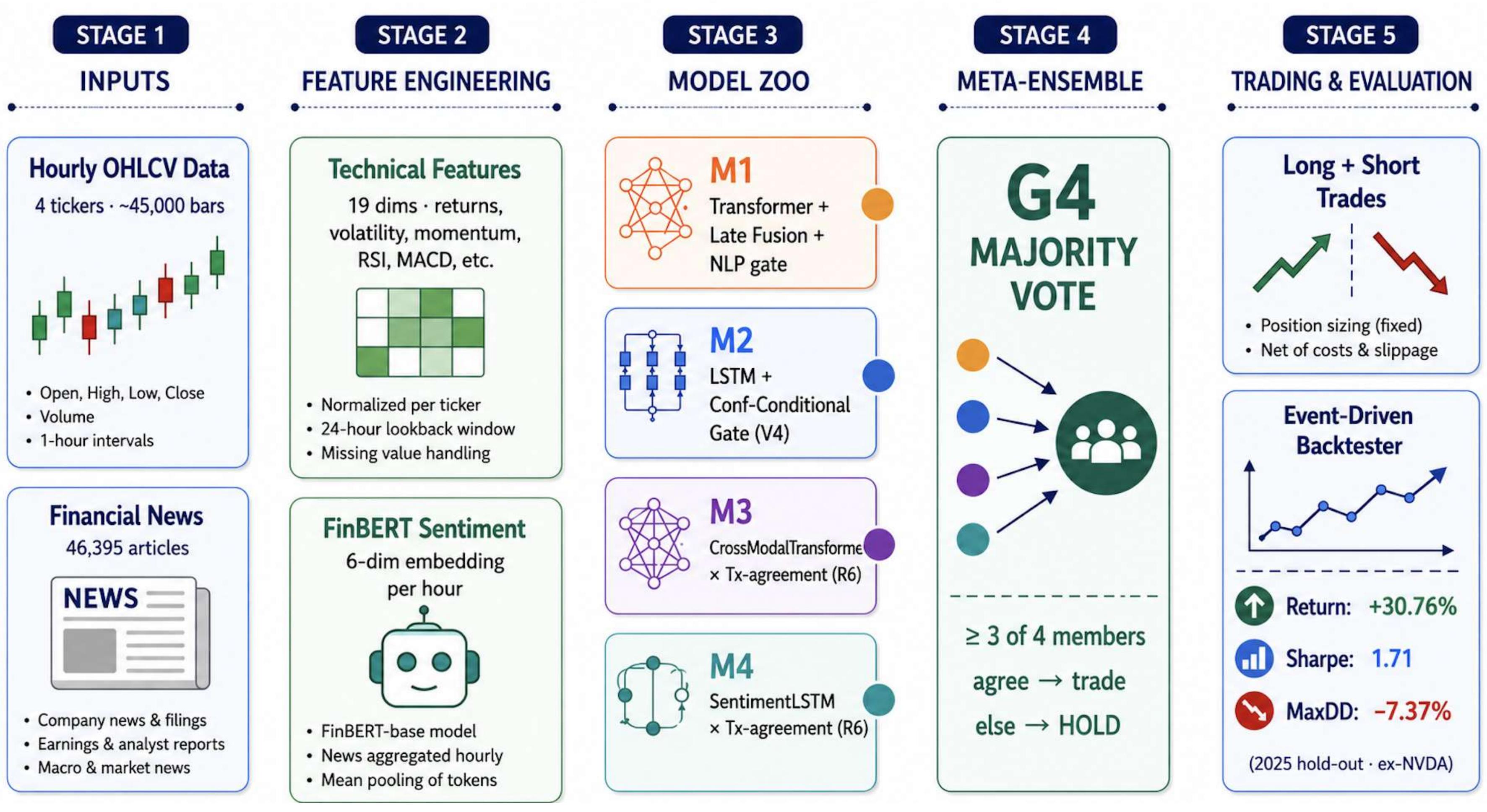
Algorithmic trading systems typically rely on either *quantitative* data (prices/volume) OR *qualitative* data (news sentiment) independently, failing to capture how markets react to both simultaneously during sudden shocks.

(H1) **NLP as a Risk Manager:** Sentiment models can act as an "emergency brake" to veto mathematically sound, but risky trades. (H2) **Meta-Ensembles:** Combining architectures that individually underperform can extract profitable signals.

**We propose FinFlow 3.0, a sentiment-augmented trading system that combines a Late-Fusion Transformer with NLP risk gating, two custom deep-fusion architectures, and a heterogeneous-member majority-vote meta-ensemble.**

## Data & Pipeline

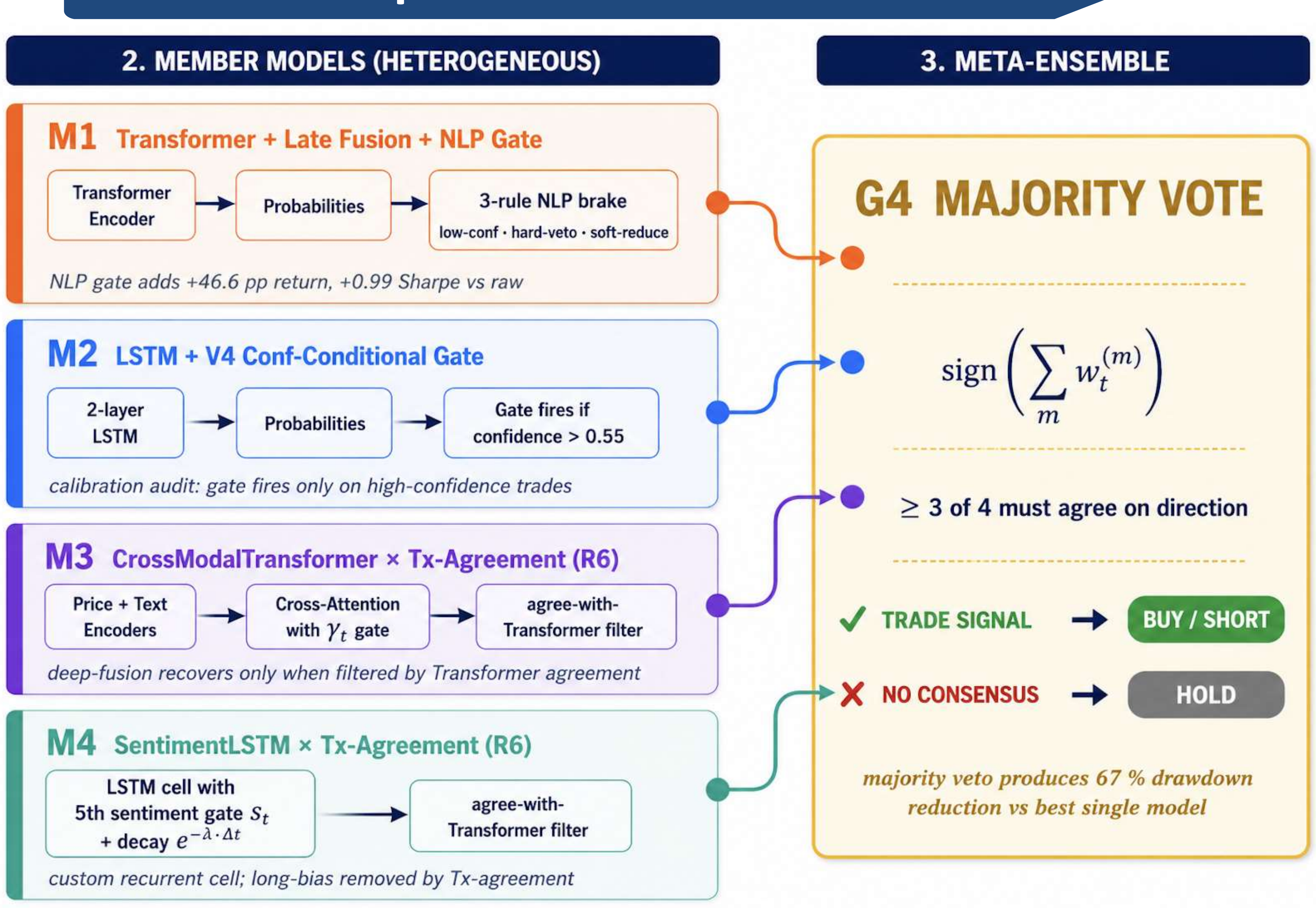
| Ticker | Sector | Role |
|---|---|---|
| NVDA | Tech (AI/GPU) | Excluded in 3.0 — idiosyncratic 2025 drawdown |
| TSLA | Consumer Discr. | Sentiment-sensitive momentum |
| JPM | Financials | Macro / rates sensitivity |
| SPY | Broad-market ETF | Benchmark and diversifier |




- **Sharpe Ratio -** risk-adjusted return per unit of volatility
- **Maximum Drawdown -** worst peak-to-trough portfolio loss
- **Profit Factor -** the ratio of gross profits to gross losses.

$$\mathrm{Sharpe} = \frac{E[r_d - r_f]}{\sigma[r_d]} \cdot \sqrt{252} \qquad \mathrm{MaxDD} = \min_t \left( \frac{V_t}{\max_{\tau \le t} V_\tau} - 1 \right)$$

## Model Development

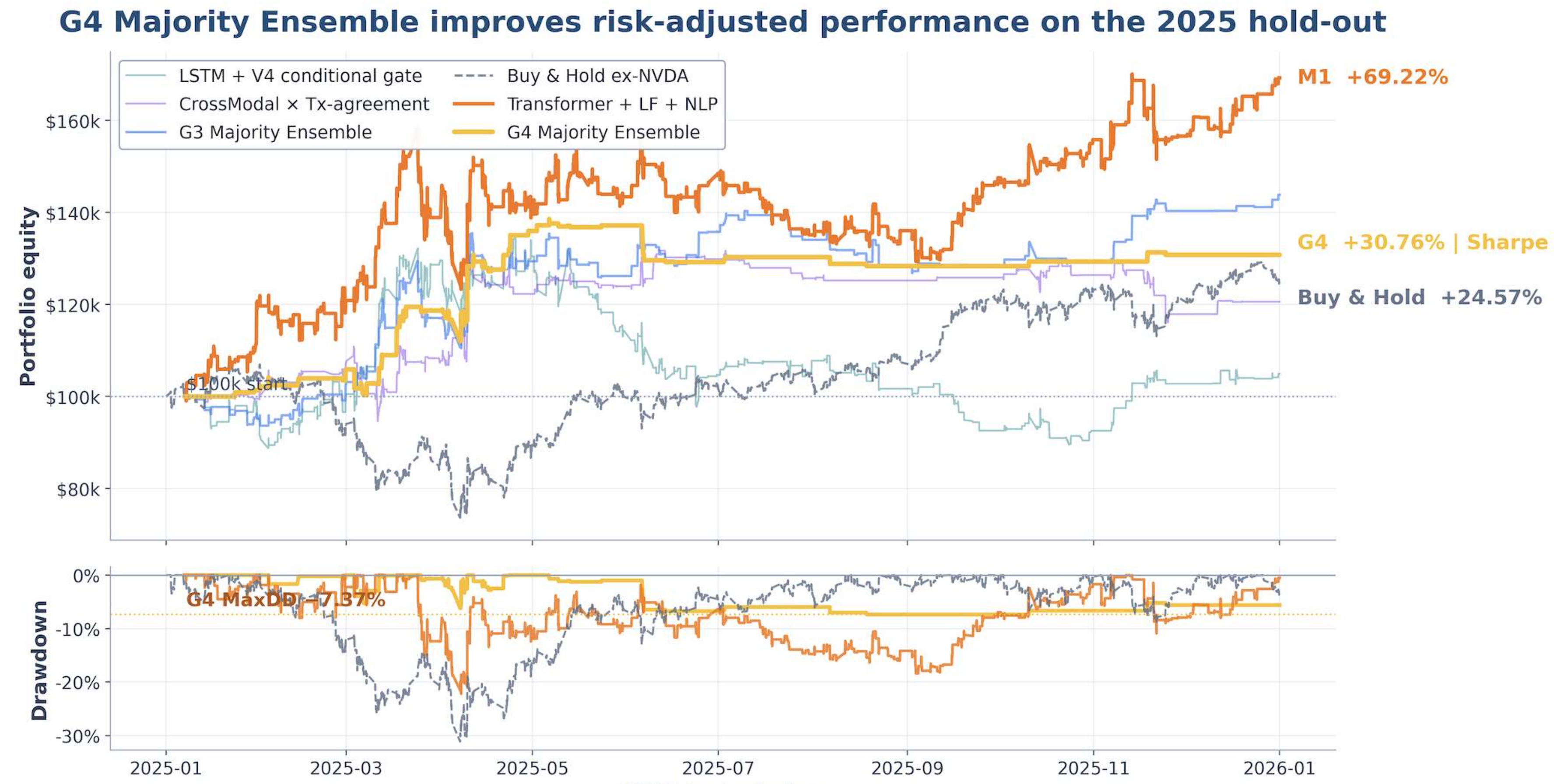



| Model | Family | What's new |
|---|---|---|
| LSTM (math-only) | Baseline 1.0 | 2-layer LSTM, 19 price features |
| Transformer (math-only) | Baseline 1.0 | 2 enc. layers, 2 heads, 64-d |
| Late Fusion + NLP gate | Rule-based | 3-rule emergency brake on Tx output |
| CrossModalTransformer | Deep Fusion 2.0 | Cross-attention price→text, gate $\gamma_t$ |
| SentimentLSTM | Deep Fusion 2.0 | Custom cell + 5th gate $s_t$, learned λ |
| G4 Majority Ensemble | Meta-ensemble 3.0 | Per-bar majority vote of M1+M2+M3+M4 |

## Results

| Strategy | Return | Sharpe | Max DD | PF | Trades |
|---|---|---|---|---|---|
| Industry Standard (Hedge Funds) | Positive | > 1.50 | > −15% | > 1.25 | – |
| G4 Majority Ensemble | +30.76 % | 1.71 | −7.37 % | 1.69 | 156 |
| Transformer + LF + NLP (M1) | +69.22 % | 1.66 | −22.21 % | 1.16 | 953 |
| G3 Majority Ensemble | +43.83 % | 1.51 | −14.65 % | 1.26 | 462 |
| Buy & Hold (ex-NVDA, benchmark) | +24.57 % | 0.80 | −31.13 % | 1.05 | 1 |
| LSTM + V4 conf-conditional gate | +4.94 % | 0.30 | −33.69 % | 1.03 | 841 |
| CrossModal × Tx-agreement (R6) | +20.58 % | 0.81 | −14.67 % | 1.14 | 453 |

**G4 Majority Ensemble improves risk-adjusted performance on the 2025 hold-out**




*Equity curves show return; drawdown panel shows capital preservation. G4 sacrifices raw return versus M1 but sharply reduces peak-to-trough risk.*

| Strategy | Return | Sharpe | Max DD | PF | Trades |
|---|---|---|---|---|---|
| G4 Majority Ensemble | +52.83 % | 2.35 | −7.36 % | 1.82 | 224 |

**Robustness check: Re-running G4 Majority Ensemble on the full 4-tickers (NVDA included)**

- **NLP as a Risk Manager (H1).** FinBERT veto on contradicting trades boosts the Transformer's return by **+46.5 pp** and Sharpe by **+0.99**.
- **Gates Require Calibration.** A blanket gate destroys the LSTM (+9.2 % → −7.4 %); restricting it to high-confidence (>0.55) signals (V4) restores profit (Sharpe +0.30). (8-variant gate sweep on val)
- **Deep-Fusion Needs Consensus.** Standalone deep-fusion fails (CrossModal 36.6 % accuracy). Requiring agreement with the Transformer (R6) lifts CrossModal to **+0.81 Sharpe**.
- **Meta-Ensembles Slash Risk (H2).** Per-bar majority vote across all four repaired members hits **Sharpe 1.71 with MaxDD −7.37 %**, a **67 %** drawdown reduction vs the best single model.

## Future Work & Limitations

- **Reinforcement Learning:** Replace fixed position sizing with an RL agent that learns dynamic allocations based on sentiment risk.
- **Learned Risk Gating:** Upgrade the rule-based NLP gate to a learned meta-controller that adapts to market volatility and regime changes.
- **Multi-Horizon Ensembling:** Expand predictions from 1-hour to daily and weekly intervals to capture cross-timescale trends.

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한양대학교 HANYANG UNIVERSITY 산업AI 인재양성 부트캠프 사업단
Industrial AI Talent Development Bootcamp Program